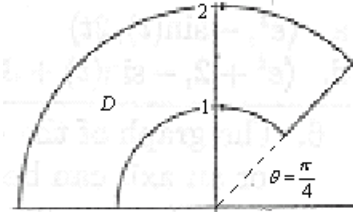


Homework Section 15.4

1. Set up an iterated integral in polar coordinates that equals $\iint_D f(x, y) dA$ if D is the region given in the following picture:



2. Sketch the region whose area is given by the integral and evaluate the integral:

$$\int_{\pi}^{2\pi} \int_1^4 r dr d\theta.$$
3. Evaluate the following integrals by converting to polar coordinates.
- $\iint_R xy dA, R = \{(x, y) \mid x^2 + y^2 \leq 4\}$
 - $\iint_R (x + y) dA, R = \{(x, y) \mid x^2 + y^2 \geq 4, x^2 + y^2 \leq 16, x \geq 0, \text{ and } y \geq 0\}$
 - $\iint_R e^{-(x^2+y^2)} dA$, where R is the region bounded by $y = -\sqrt{9-x^2}$ and the x -axis.
 - $\int_0^1 \int_0^{\sqrt{1-x^2}} e^{x^2+y^2} dy dx$
4. Use a double integral in polar coordinates to find the area taken up by one petal of the rose curve $r = \sin 2\theta$.
5. Find the volume:
- Under $z = x^2 + y^2$ and above the region $R = \{(x, y) \mid x^2 + y^2 \leq 4\}$.
 - Inside the sphere $x^2 + y^2 + z^2 = 16$ and outside the cylinder $x^2 + y^2 = 4$.
 - Above the cone $z = \sqrt{x^2 + y^2}$ and below the sphere $x^2 + y^2 + z^2 = 1$.